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Oefeningenreeks 1B nr. 37.4

$-3+4i$

$= (x+yi)^2 = x^2 - y^2 + 2xyi$

$$\begin{cases} x^2 - y^2 & = & -3 \\ 2xy & = & 4 \end{cases}$$

$$\begin{cases} x^2 - y^2 & = & -3 \\ xy & = & 2 \end{cases}$$

$$\Leftrightarrow \begin{cases} x^2 - y^2 & = & -3 \\ x^2 \cdot (-y^2) & = & -4 \\ xy > 0 \end{cases}$$

$\lambda^2 + 3\lambda - 4 = 0$

$D = b^2 + 4ac = 9 + 16 = 25$

$\lambda_1 = \dfrac{-b-\sqrt{D}}{2a} = -4$

$\lambda_2 = \dfrac{-b+\sqrt{D}}{2a} = 1$

$x^2 = 1$

$-y^2 = -4$

$x = \sqrt{1}$ en $y = \sqrt{4}$

Oplossing: $1+2i$ of $-1-2i$

References